



PHENOTYPE/ECOSYSTEM INTERACTION

Alexandros Kourtidis & Kristina Yefimak



Alexandros (Alex)



Biology BSc @ Athens, Greece

Erasmus+ studies @ Grenoble, France

Evolution, Ecology and Systematics MSc
@ Munich, Germany

PhD in Evolutionary Genomics of
Daphnia magna @ Leuven, Belgium

Singing + Dancing + Acting + Alex = BFF <3

Kristina



Bioscience engineering



KULAK



PhD on microbiome *Daphnia magna*



Gym, reading

Levels of organisation and diversity of life

Examples

Diversity

• Molecules	DNA, proteins, RNA	mRNA, tRNA, rRNA
• Organelles	Mitochondria, golgi, Chloroplasts	Different chloroplast types i.e. amyloplast
• Cells – Tissues	Muscle, neuron, bone	Different neuron types i.e. motorneuron
• Organs	Heart, brain, bones	Different bone types i.e. flat, long
• Organisms	Animals, plants, fungi, bacteria	Animal kingdom
• Populations	Daphnia, elephants	Different lake/island populations of 1 species
• Communities	Forest, pond, dessert communities	Different forest communities
• Ecosystems communities	Forest, pond, dessert ecosystems	Different types of forest with other living
• Biosphere		

Levels of biodiversity



Population level



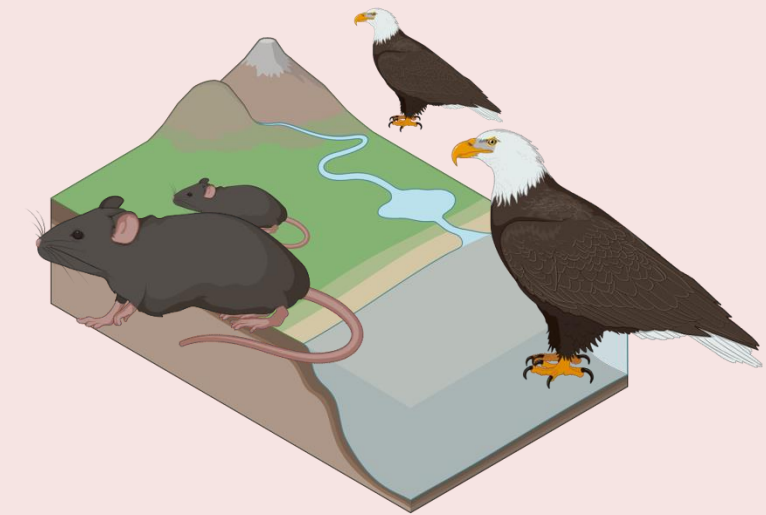
Community level

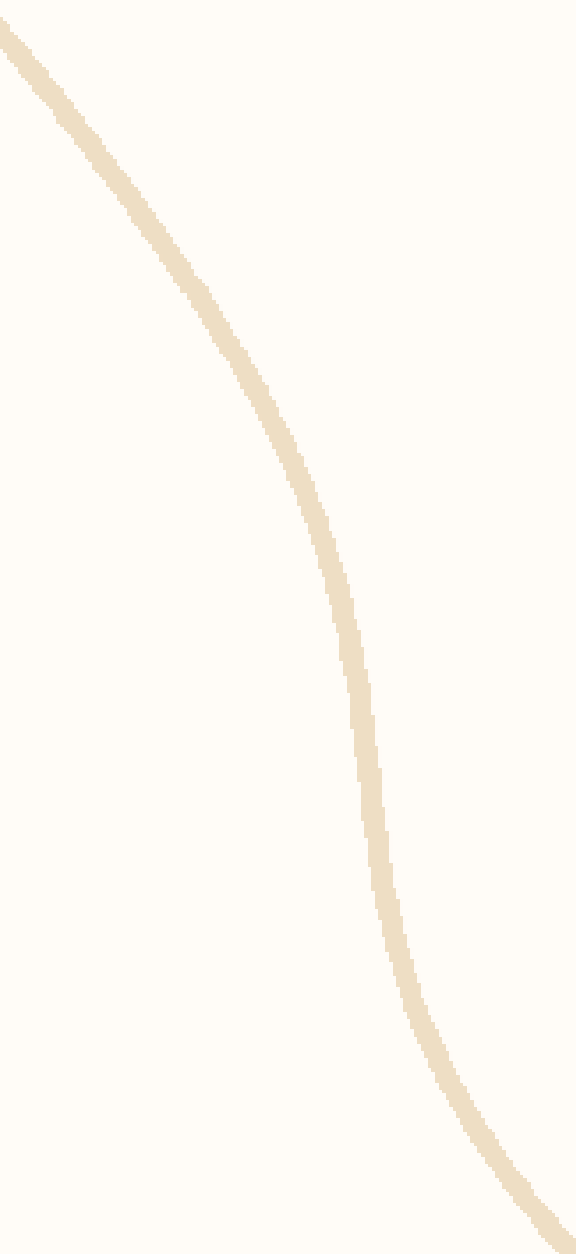


Ecosystem level

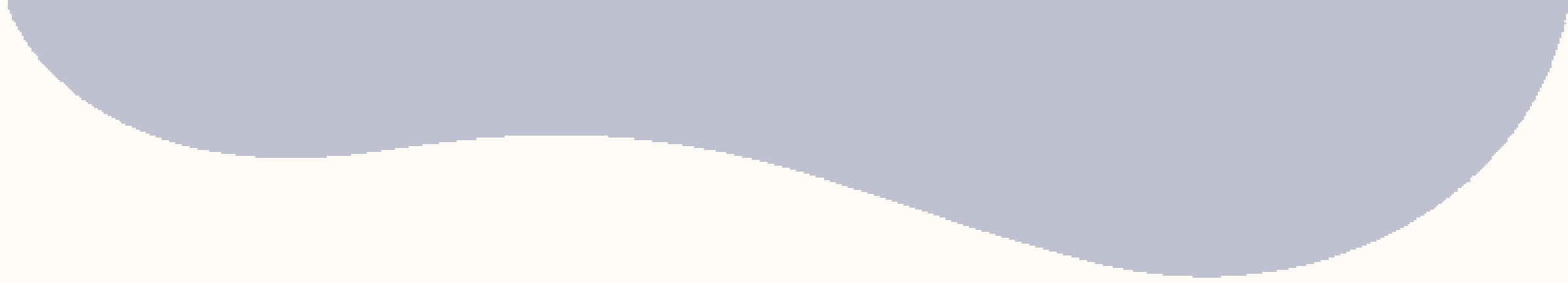


Interconnected levels!





Biotic interactions



Biotic interactions= Interactions
between living organisms in an
ecosystem.

Interactions can be competitive,
cooperative, exploitative, or neutral.

Biotic interactions shape population
dynamics, community structure, and
evolutionary processes.



1) Competition (-/-)

Two species compete for the same resource

-Lions and hyenas

Interspecific competition is between different species

Intraspecific competition is between members of the same species

-Two male deer competing for mates

Natural selection?

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Intraspecific vs Interspecific Impacts



Intraspecific (within species)

Drives adaptive
evolution

Directional selection
and evolutionary
change within
populations

Interspecific (between species)

Shapes community
structure and
diversity

Drive coevolution and
species
diversification

2) Mutualism (+/+)

Both benefit

- Bees pollinate flowers
- Smaller fish cleaning bigger fish
- Gut bacteria in humans

Obligate: Can't survive without each other

Facultative: Helpful but not essential

Evolutionary impact: Can lead to coevolution, specialization, and symbiosis

3) Commensalism (+/0)



One benefits, the other is unaffected

-Barnacles on whales

-Birds nesting in trees

4) Amensalism (0/−)



One is harmed, the other unaffected

-Large trees block sunlight to small plants

Not intentional

5) Parasitism (+/-)



Parasite lives on/in host, harming it

-Ticks on dogs



Often highly specialized →
coevolutionary dynamics.



Affects host survival, reproduction,
and behavior → Can regulate host
populations (density dependence)



6) Predation (+/-)



One species eats another

-Fox and rabbit



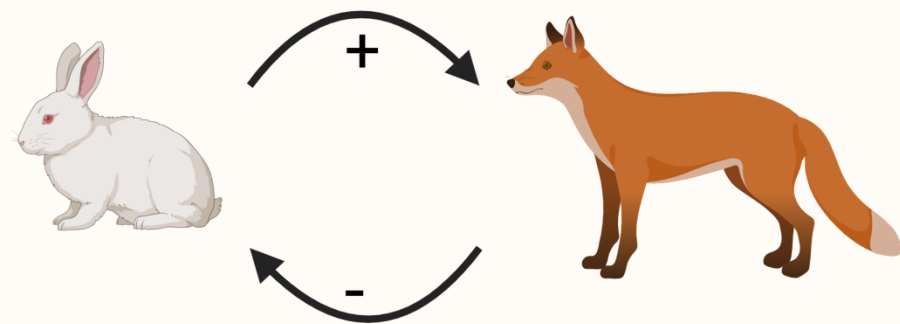
Controls prey populations



Lotka-Volterra Model



Lotka-Volterra



Describes how the population sizes of predators and prey change over time

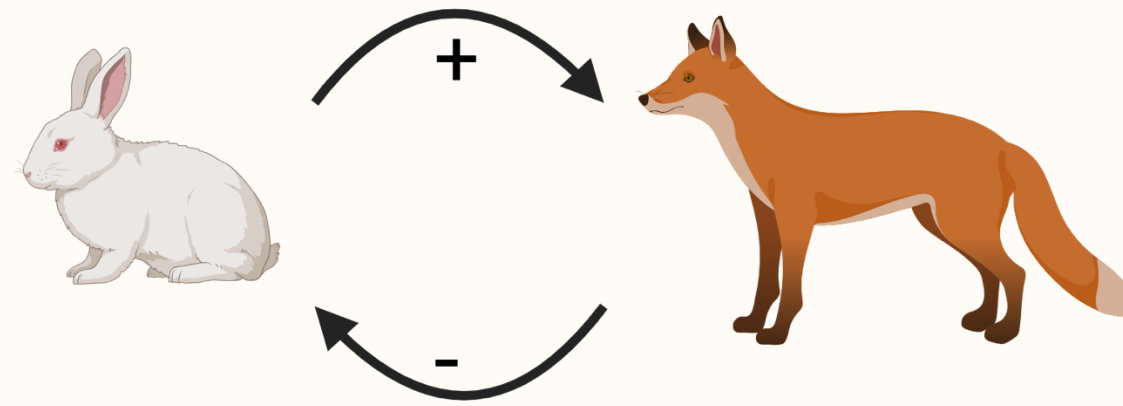
Predators need prey to survive. Prey are limited by predators. This feedback leads to cycles

Shows that each species' growth depends on the other

INTUITION

- Prey grow fast, unless eaten.
- More prey → more predators.
- More predators → prey decline → predator starvation → cycle restarts.

Prey population
growth:
 $dN/dt = rN - aNP$

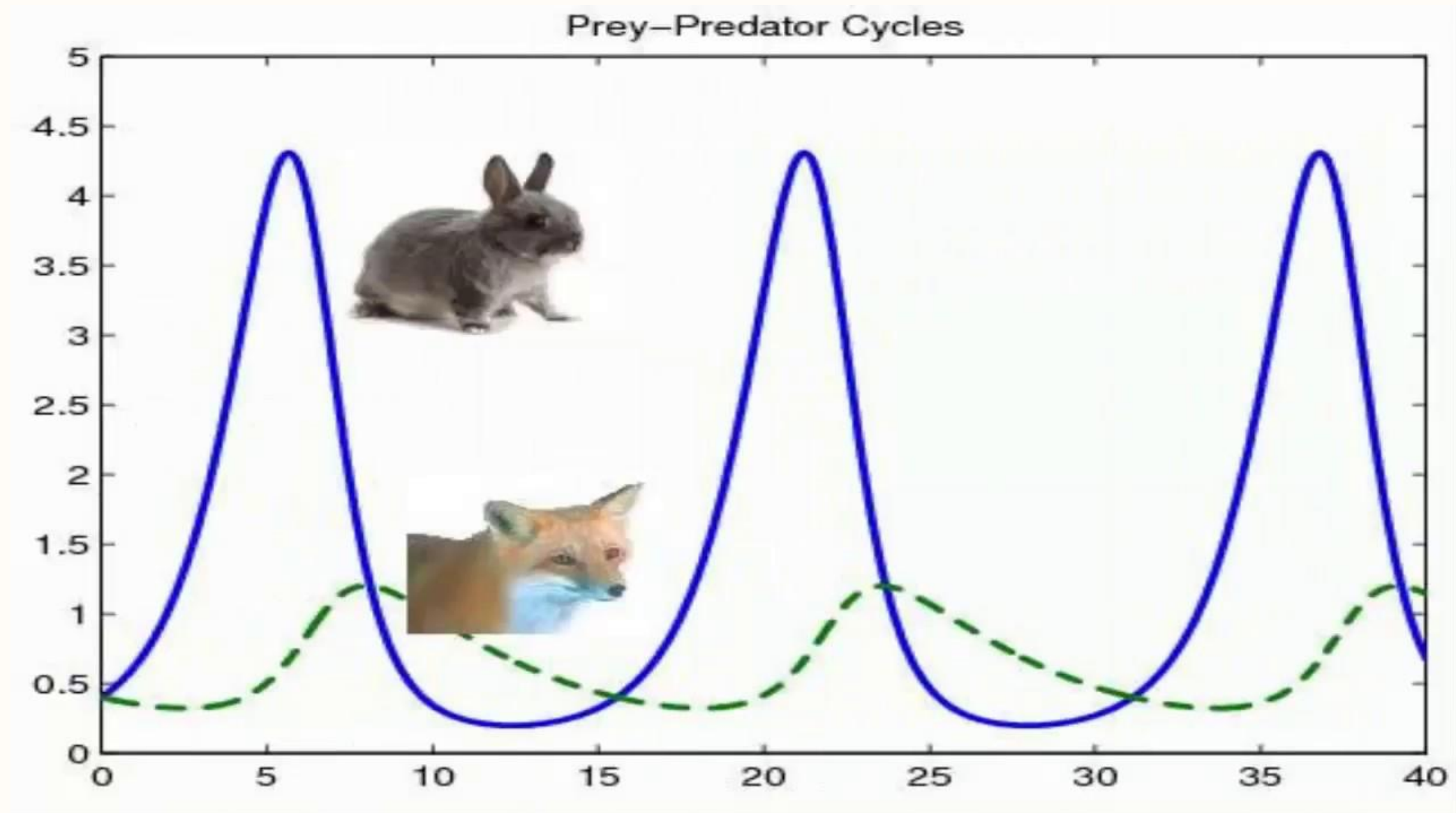
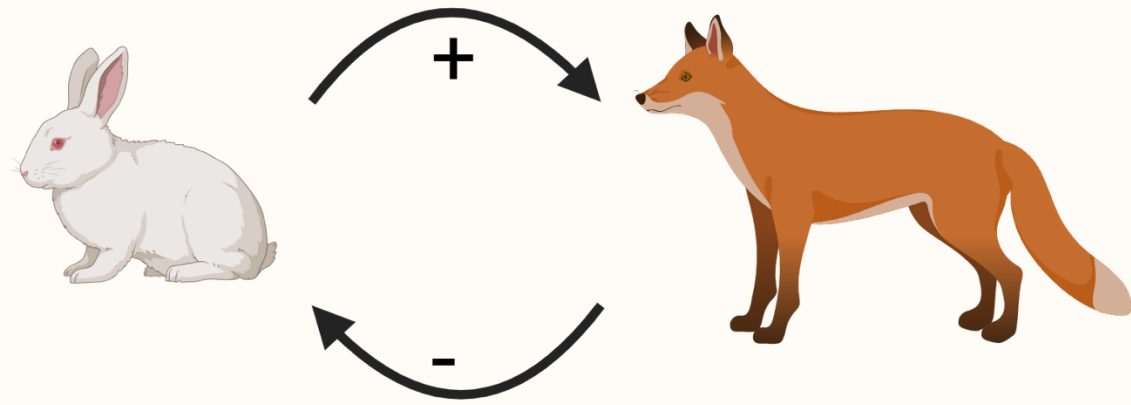


Predator
population growth:
 $dP/dt = baNP - mP$

- rN : exponential prey growth in absence of predators = 🐇 boom
- aNP : rate at which predators eat prey = 🦊 eating 🐇

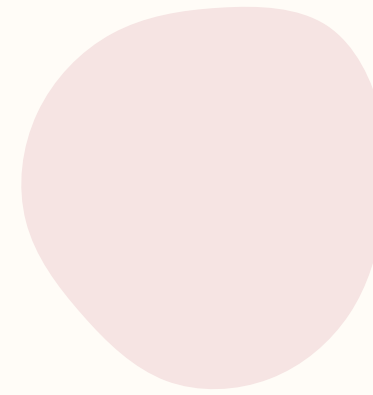
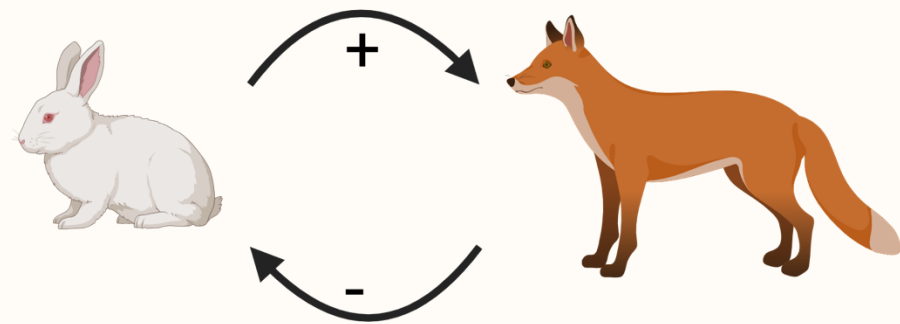
- N = number of prey
- P = number of predators

- $baNP$: predators gain energy from prey
- mP : predator deaths (without enough food) = 🦊 starves



- Prey peak → predator rise.
- Predator peak → prey crash.
- Predator crash → prey rebound.

Lotka-Volterra: Realistic?



Predator depends
only on this one
prey



No environment
carrying capacity



No time delays or
evolution

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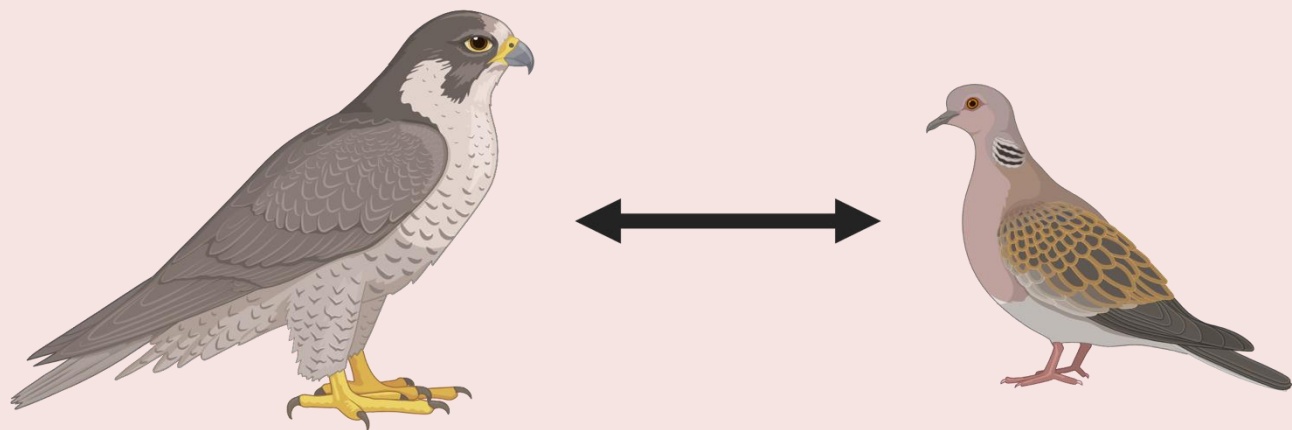
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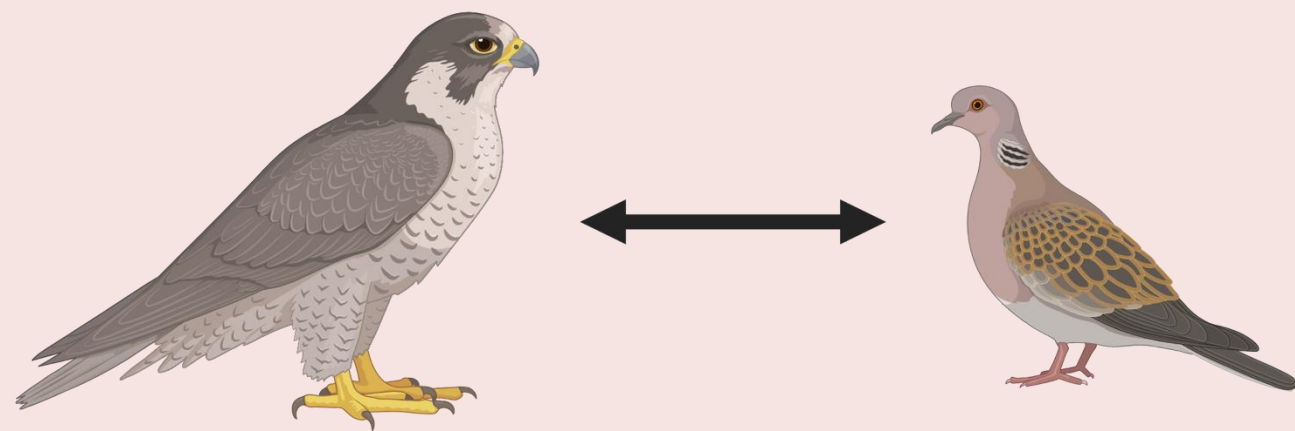
Natural selection?

Game Theory and Evolutionary Stable Strategies: HAWK vs DOVE



- Models contests over resources (food, territory, mates)
- Each strategy has costs and benefits
- Which strategy is better depends on how common each one is
- This models frequency-dependent selection

HAWK vs DOVE



	OPPONENT HAWK	OPPONENT DOVE
HAWK (player)	-25	+50
DOVE (player)	0	+25

Natural selection balances aggression with cost of injury

HAWK vs DOVE

round	strategy	payoff
1	hawk	50
1	dove	0
1	hawk	50
1	dove	0
1	dove	25
1	dove	25
1	dove	25
1	dove	25
1	hawk	50
1	dove	0
1	dove	25
1	dove	25
1	dove	25
1	dove	25

strategy	total
Hawk(3)=22%	150/3=50
Dove(11) =78%	200/11=18

Natural selection balances aggression with cost of injury

HAWK vs DOVE

round	strategy	payoff
2	dove	25
2	dove	25
2	dove	25
2	dove	25
2	dove	25
2	dove	25
2	hawk	-25
2	hawk	-25
2	dove	25
2	dove	25
2	hawk	50
2	dove	0
2	dove	25
2	dove	25

strategy	total
Hawk(3)	Sum/14=0
Dove (11)	250/12=22

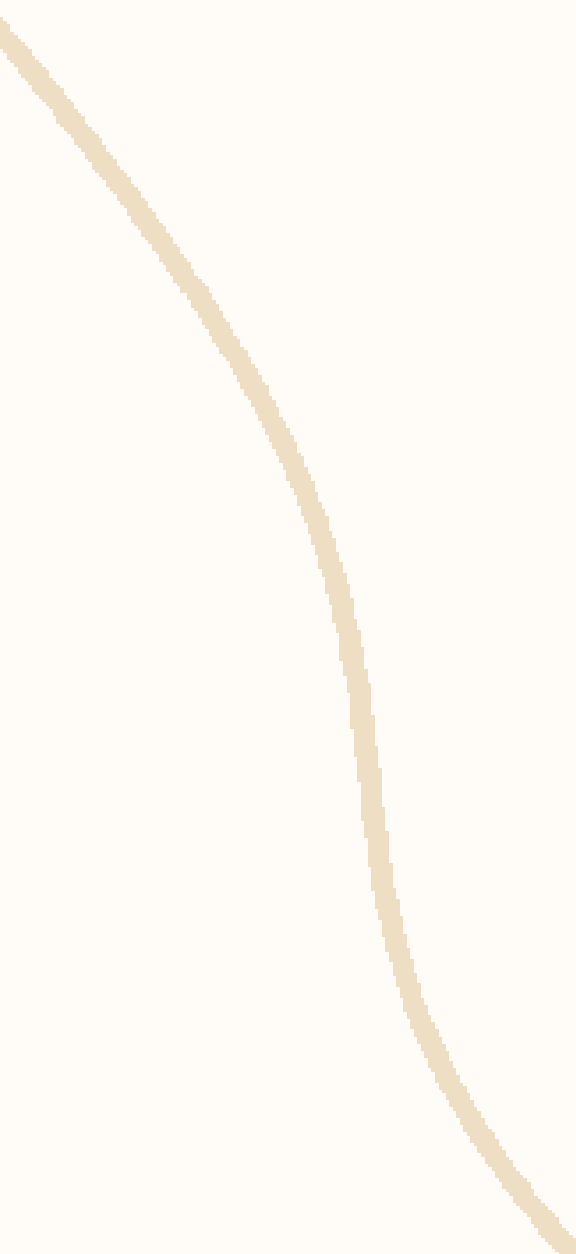
Natural selection balances aggression with cost of injury

HAWK vs DOVE

round	strategy	payoff
3	hawk	50
3	dove	0
3	dove	25
3	dove	25
3	dove	25
3	dove	25
3	hawk	-25
3	hawk	-25
3	hawk	50
3	dove	0
3	Hawk	50
3	dove	0
3	hawk	50
3	dove	0

strategy	total
Hawk(6)	$200/6=33.3$
Dove(9)	$100/9=11.1$

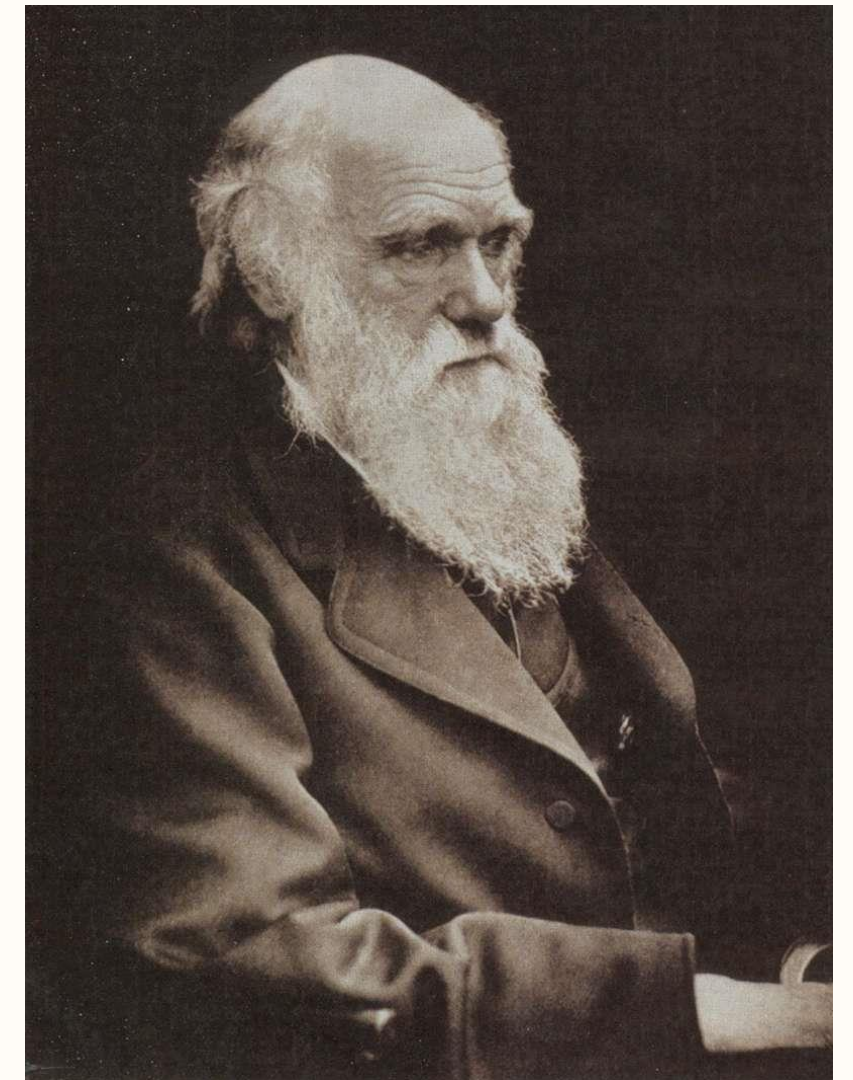
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Introduction to selection

Darwin

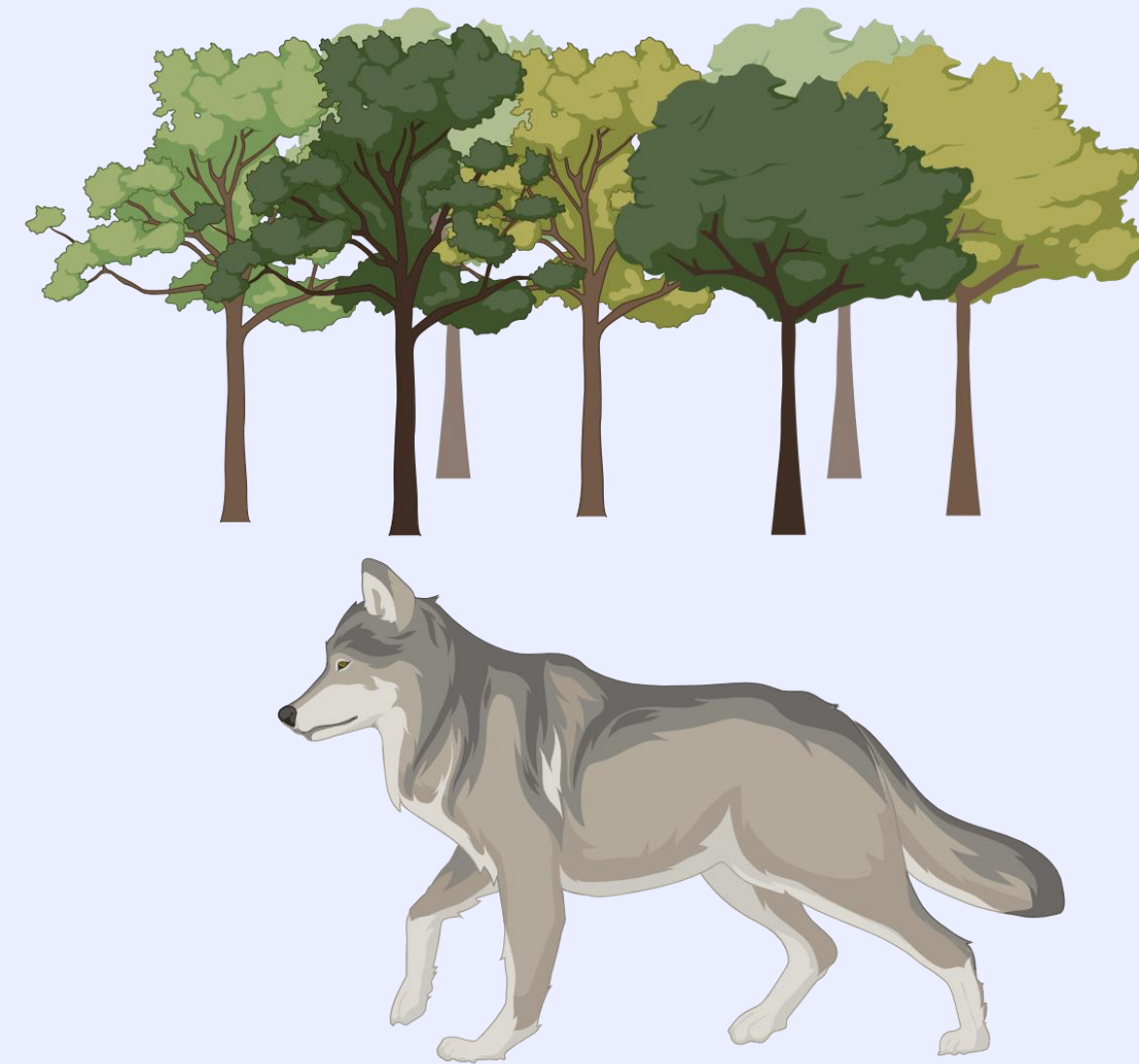
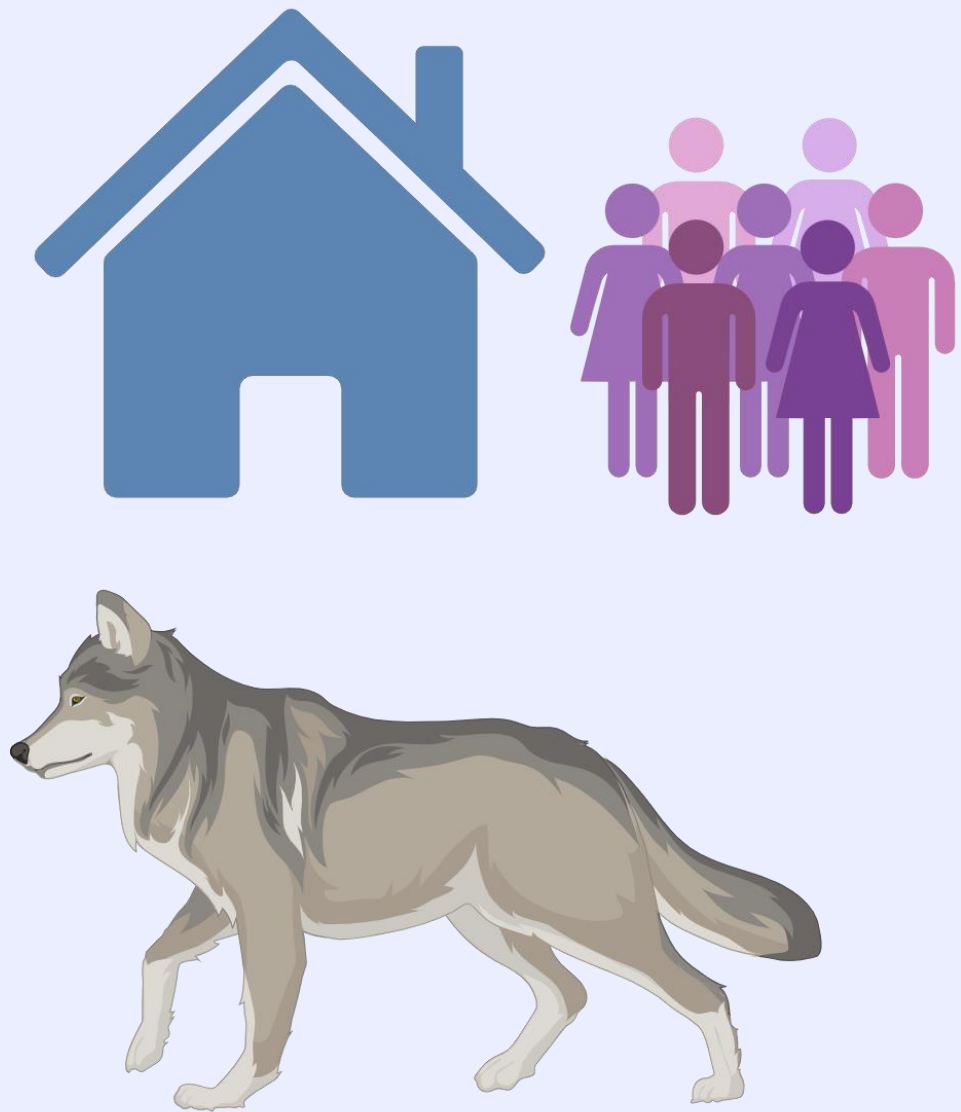
He is best known for developing the theory of evolution by natural selection.



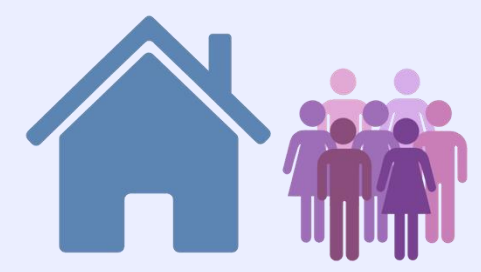
If you had a
bunch of wolves
and wanted a
Chihuahua, how
would you
create one?



Answers



Traits don't evolve because of need or use/disuse



Natural selection



There are differences (e.g. wolf pups vary in size)



The variation is genetic (small parents → small pups).



Some individuals reproduce more (breeder chooses smaller pups to reproduce).

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Natural selection?

Red queen dynamics



organisms must constantly evolve just to keep up (e.g. *Daphnia* and parasites)

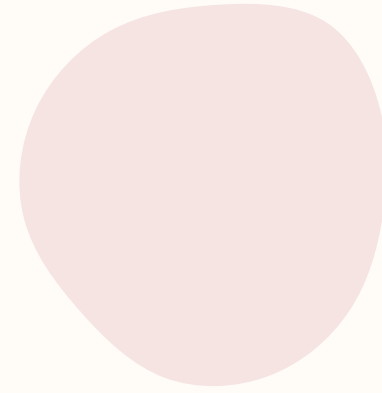
This is natural selection in a feedback loop — your environment is evolving too

Host evolves better defenses → parasites evolves to overcome them, but infection rate often stays the same

Alice in wonderland: Red queen: "It takes all the running you can do to stay in the same place."

- Evolution is not about becoming "perfect", it's about keeping up.
- Species are interconnected, as one evolves, others must respond.
- Evolution is ongoing and dynamic, not a one-time event

Antibiotic resistance



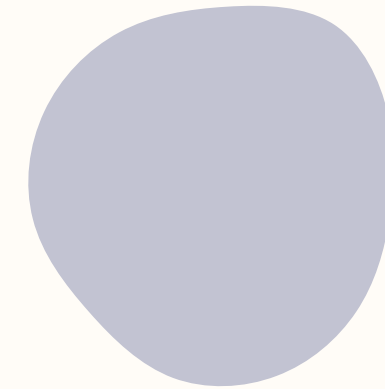
Antibiotics = strong
selective pressure



Resistant mutants
survive, reproduce
→ population quickly
shifts



New antibiotics →
new resistance
mechanisms → new
drugs...



Natural selection
1) some bacteria are resistant
1) resistance genes are passed on
2) resistant cells survive and reproduce



Ready for some labwork?

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